INCH-POUND

MIL-C-82768
5 June 1989

# MILITARY SPECIFICATION

# CUTTER, CARTRIDGE ACTUATED, TCU-4/E

**This specification is approved for use by all Departments and Agencies of the Department of Defense.**

## 1. SCOPE

1.1 **Scope.** This specification covers the minimum requirements for the procurement of the TCU-4/E Cartridge Actuated Cutter. Cutters may be procured in either of two configurations.

1.1.1 **Combination of adopted items kit for the TCU-4/E.** The TCU-4/E when purchased in kit form consists of an unassembled TCU-4/E and contains all hardware necessary for installing the TCU-4/E onto the M58/68 Linear Demolition Charge System (LDCS).

1.2.1 **TCU-4/E cartridge actuated cutter.** The TCU-4/E when purchased alone comes as an assembled cutter and is ready for testing or installation onto an existing M58/68 LDCS equipped with TCU-4/E mounting hardware.

## 2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1377

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-D-23615 | Design and Evaluation of Cartridge Actuated Devices |
| MIL-I-81557 | Ignition Element, Electric, Mark 17 Mod 0 |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-129 | Marking for Shipment and Storage |
| MIL-STD-453 | Inspection, Radiographic |
| MIL-STD-810 | Environmental Test Methods and Engineering Guidelines |

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

**2.1.2 Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

Naval Air Systems Command (CAGE Code 30003)

| | |
|---|---|
| DL613468 | Ignition Element, Electric, MK 17 Mod 0 |

Naval Sea Systems Command (CAGE Code 53711)

| | |
|---|---|
| DL6261221 | Cutter, Cartridge Actuated, TCU-4/E |
| 6261288 | Packaging for the TCU-4/E Cartridge Actuated Cutter |

PUBLICATIONS

Naval Ordnance Station, Indian Head (CAGE Code 14803)

| | |
|---|---|
| WS 25712 | Rope, Nylon, With Three Electrical Conductors |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

(Application for copies should be addressed to: Commanding Officer, Naval Ordnance Station, (ATTN: Code 802), Louisville, KY 40214-5001).

**2.2 Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

**3.1 First article.** When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.5). The MK 17 MOD 0 Ignition Elements used in the cutters shall be furnished from lots which were accepted in accordance with MIL-I-81577.

**3.2 Materials.** All materials, parts, and assemblies shall be in accordance with DL6261221, this specification, and the documents referenced herein.

**3.2.1 MK 17 MOD 0 Ignition Elements.** Only MK 17 MOD 0 Ignition Elements manufactured in accordance with the drawings listed on DL613468 within the previous six months shall be used in the cutters for testing or delivery.

**3.3 Performance and product.** Cutter shall meet the requirements of DL6261221 and shall be capable of cutting completely an 11/16 inch cable as described in WS 25712 when functioned. Cutter shall cut cable completely within 50 msec.

**3.3.1 Visual.** Cutters, when subjected to the visual examination of 4.7.1, shall meet all requirements of the drawings. Cutters shall be free of the following visible defects: chips, burrs, rough edges, incomplete threads, or any other imperfections which could harmfully affect either assembly or functioning.

**3.3.2 Radiographic.** Cutters shall be free of the following defects when examined radiographically as described in 4.7.2: improper assembly of components, charges or inert components missing, or any other imperfections which could adversely affect performance.

**3.3.3 Bridgewire resistance.** The resistance of the bridgewire in each cutter, when measured as specified in 4.7.3, shall be not less than 1.0 ohm nor more than 1.2 ohms.

**3.3.4 Six foot drop.** Cutters shall be safe for handling and shall function as designed after being subjected to the test of 4.7.4.

**3.3.5 Shock.** Cutters shall be safe for handling and shall function as designed after being subjected to the test of 4.7.5.

**3.3.6 Vibration.** Cutters shall be safe for handling and shall function as designed after being subjected to the test of 4.7.6.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

**3.3.7 Temperature and humidity.** Cutters shall be safe for handling and shall function as designed after being subjected to the test of 4.7.7.

**3.3.8 Environmental.** After being subjected to the tests of 4.7.4 through 4.7.7, cutters shall be free of the following defects when examined radiographically as described in 4.7.2: improper assembly of components, charges or inert components missing, or any other imperfections which could adversely affect performance.

**3.4 Workmanship.** The workmanship shall be that required by the applicable drawings, referenced specifications, and the best industrial practices governing the quality production of interchangeable parts for maintaining the dimensions, finishes, tolerances, and quality specified herein. All components and assemblies shall be free from burrs, contamination, corrosion, chips, incomplete threads, sharp edges, or foreign material which could affect assembly, result in malfunction of the unit or be a safety hazard in handling.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Responsibility for inspection.** Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

**4.1.1 Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

**4.2 Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.5).

b. Quality conformance inspection (see 4.6).



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

**4.3 Inspection conditions.** Unless otherwise specified, all inspections shall be performed under the following conditions:

a. Temperature: Room ambient 65 to 95°F
b. Altitude: Sea level
c. Atmospheric pressure: 14.7 psia
d. Humidity: Room ambient to 95 percent relative, maximum

**4.4 MK 17 MOD 0 Ignition Elements.** Only MK 17 MOD 0 Ignition Elements from an accepted lot shall be used in production/installation of the cutter.

**4.5 First article inspection.** At the beginning of regular production, the contractor shall submit an initial production sample of 45 cutters to a Government approved facility for evaluation. The sample shall be manufactured in the same manner, using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, must be the same as will be used for regular production and shall be obtained from the same source of supply. After inspection and provisional acceptance at source, the first article sample will be inspected for all requirements of the drawings and specifications and will be subjected to first article tests as described in table I. The Government reserves the right to require new first article samples until such time as an acceptable sample is submitted (see 6.3).

TABLE I. First article test requirements.

| Test | Quantity (45 total) | | | | | Paragraph |
|---|---|---|---|---|---|---|
| Group | 1 | 2 | 3 | 4 | 5 | |
| Visual | 3 | 12 | 12 | 12 | 6 | 4.7.1 |
| Radiographic | 3 | 12 | 12 | 12 | 6 | 4.7.2 |
| Resistance | 3 | 12 | 12 | 12 | 6 | 4.7.3 |
| Six foot drop | 3 | | | | | 4.7.4 |
| Shock | | 12 | | | | 4.7.5 |
| Vibration | | | 12 | | | 4.7.6 |
| Temperature and humidity | | | | 12 | | 4.7.7 |
| Radiographic | 3 | 12 | 12 | 12 | | 4.7.2 |
| Resistance | 3 | 12 | 12 | 12 | | 4.7.3 |
| Functional | | | | | | 4.7.8 |
| -65°F | | 4 | 4 | 4 | | |
| 70°F | 3 | 4 | 4 | 4 | 6 | |
| 200°F | | 4 | 4 | 4 | | |



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

**4.6 Quality conformance inspection.** Quality conformance inspection shall include the examinations and tests of 4.6.1 through 4.6.5. The MK 17 MOD 0 Ignition Elements used in the cutters shall be furnished from lots which were accepted in accordance with MIL-I-81577.

**4.6.1 Production lot.** The suggested cutter production lot size is 250. The maximum lot size is 500. Only MK 17 MOD 0 Ignition Elements from a single lot shall be used in a production lot of cutters. One production lot of MK 17 MOD 0 Ignition Elements can be used in more than one lot of cutters.

**4.6.2 Lot acceptance testing.** All cutters within a production lot shall be subjected to the visual inspection of 4.7.1. After completion a sample of cutters shall be selected for lot acceptance testing. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a contractor shall be in accordance with MIL-STD-105, inspection level 1. Cutters shall be selected at random from the regular production lot.

**4.6.3 Resistance testing.** The bridgewire resistance shall be tested as specified in 4.7.3.

**4.6.4 Functional testing.** Units shall be tested as specified in 4.7.8 except that units shall be divided equally between the two temperature groups, -65°F or 200°F, with any remaining units put in the 200°F group.

**4.6.5 Inspection of packaging.** Packaging and marking shall be inspected for conformance to section 5.

**4.7 Methods of inspection.**

**4.7.1 Visual examination.** All test units shall be visually inspected for defects in material or workmanship and compliance with applicable drawings and documents. Each assembly shall be labeled with a control number. Verification of successful numbers shall be consecutive, beginning with 001.

**4.7.2 Radiographic examination.** Assembled sample cutters shall be radiographically examined in accordance with MIL-STD-453, with the exception that a cutter assembly will be used for the penetrameter. Any observable imperfections in components or assembly shall be cause for rejection. The cutters shall be positioned for the most revealing exposure, with the long axis of each assembly perpendicular to the rays of the X-ray machine. All cutters shall be identified with serial numbers prior to X-ray examination. The serial numbers shall be in consecutive numerical order beginning with the number 001 in each production lot. The cutters shall be arranged on boards or trays in consecutive order and each radiograph shall carry a permanent identification of the units displayed thereon.

**4.7.3 Resistance.** The bridgewire resistance shall be measured and recorded with a test meter having an accuracy of $\pm$ 0.05 ohms with an impressed



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

current limited to 0.10 amperes. Only ignition elements having a resistance of 1.0 to 1.2 ohms shall be acceptable.

**4.7.4 Six foot drop.** The sample cutters shall be dropped from six feet onto a two inch thick, minimum, steel plate embedded in concrete along each of the three mutually perpendicular axes. A guidance system shall be used to ensure the proper impact orientation. Dropping shall not prematurely function units, dud, nor induce any loose parts.

**4.7.5 Shock.** Units shall be shock tested to 20g following the methods of MIL-STD-810B, procedure 1 (terminal peak sawtooth, 11 ms), 2 units per axis. Units shall be shocked in the "+" and "-" x direction and in the "+" direction in the other axes. Units shall then be resistance tested, x-rayed, and functioned at 70°F. When functionally tested per paragraph 4.7.8, units shall meet the requirements of paragraph 3.3. Shock shall not prematurely function units, dud, or induce any loose parts.

**4.7.6 Vibration.** Vibration tests shall be done on each test sample along the three mutually perpendicular axes as follows. The sample units shall be divided evenly among the three conditioning temperatures. A 5 to 2000 Hz resonance search shall be done on the unit at each orientation while conditioned at -65, 70, or 200°F. Resonance search shall be done at an applied acceleration of one g for a total of nine searches (3 temperatures by 3 axes). In order to reduce future conditioning time, the second unit may also be installed during resonance search but need not be instrumented. Mount accelerometer near the middle of the cutter's longest side. If one or two resonant frequencies are found (if more than two, use the two most severe), each unit shall be vibrated at it's resonance frequency for twenty minutes, or ten minutes at each of two resonant frequencies, in each applicable axis temperature combination. If no resonant frequencies are found, each unit shall be vibrated for one hour at -65°F, one hour at 70°F, and one hour at 200°F at each of the three mutually perpendicular axes (reduce these times by the amount of resonance vibration time if resonant frequencies are found), by cycling the frequency logarithmically from 5 to 2000 Hz. Vibration (exclusive of resonance searches) shall be performed at an applied double amplitude of 0.1 inch (5 to 20 Hz) or an applied acceleration of +2 g (20 to 2000 Hz) whichever results in the lower value of acceleration (see curve b of figure 514.2-2 of MIL-STD-810C). Vibration shall not prematurely function units or induce any loose parts.

**4.7.7 Temperature and humidity cycling.** The samples shall undergo temperature and humidity cycling and storage tests in accordance with paragraph 4.7.4 of MIL-D-23615B. Units shall be resistance tested and x-rayed and shall meet the design performance requirements of paragraph 3.3 when functionally tested per paragraph 4.7.8.

**4.7.8 Functional.** Cutters shall be functioned at -65°F, 70°F, or 200°F. Prior to temperature conditioning, cutters shall be fitted with an 8 $\pm$ 1 inch cable segment with a breaklead installed between the cable and the anvil. The breaklead shall be used to measure the actuation time of the cutter. Cutters shall then be conditioned at temperature for a period not less than four hours



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

and not greater than 24 hours. Units shall be removed from conditioning and then be functioned within five minutes. No tensile load is to be applied to the cable during functioning. Cutters shall be functioned by supplying a current of 4.0 amperes to the two pins of the MK 17 MOD 0 Ignition Element. In the event that a cutter is out of conditioning for more than five minutes, the cutter shall be allowed to remain at 70°F for at least four hours and then be reconditioned as described above.

## 5. PACKAGING

**5.1 Packaging and marking.** All cutters and cutter components shall be packaged and marked as described in Drawing 6261288 and MIL-STD-129.

**5.2 Packaging for immediate use.** Commercial preservation and packing, with the requirements of Drawing 6261288 included, shall be used for packaging of all cutters which are intended for expenditure in first article and production lot acceptance tests.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory).

**6.1 Intended use.** The TCU-4/E Cartridge Actuated Cutter is intended for use on the M58/68 Linear Demolition Charge System as a backup means of separating the line charge from the deploying vehicle.

**6.2 Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 4.7.5 and 4.7.7).

c. Whether first article inspection is required and, if so, specify the test activity (see 4.5).

d. Production lot sample size and sampling plan (see 4.6.2).

e. Items of data required for each first article and production lot (see 4.5 and 4.6).

f. Level of packaging and packing desired (see 5.1).

**6.3 First article.** When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.5 The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

**6.3.1 Waiver of first article test.** First article tests should not be waived when one of these three conditions exists:

1) The contractor has never manufactured this cutter.
2) Elapsed time from date of last acceptable delivery exceeds 18 months.
3) Contractor has made changes in location of assembly plant or has experienced a complete change-over of production personnel.

**6.4 Explanation of terms.** For the purposes of this specification, NAVSUPINST 4030.28A (20 June 1975) paragraphs 1-4, 2-2, 2-3 and Appendix A, when applicable shall be used to clarify terms.

**6.5 Safety precautions.** The safety precaution requirements of the "Contractors' Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DoD 4145.26M) are applicable and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) 23.2.

**6.6 Subject term (key word) listing.**

Demolition
Ignition element
Linear Demolition Charge System (LDCS)

Custodian:
Navy - OS
Army - ME

Preparing activity:
Navy-OS

Review activity:
Navy - MC

(Project 1377-0953)



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.

# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-C-82768 | CUTTER, CARTRIDGE ACTUATED, TCU-4/E |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)* _____ |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

*(TO DETACH THIS FORM, CUT ALONG THIS LINE.)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:41Z
Check the source to verify that this is the current version before use.